Global Breast Cancer Outcomes Analysis: Evaluating Early Detection, Treatment Access, and Survival Inequality in Healthcare Systems

1. Executive Summary

Breast cancer outcomes vary significantly across countries, driven by disparities in early detection, treatment access, and healthcare infrastructure. This analysis combines global country-level data, stage-based survival rates, and risk factor insights to identify the key drivers behind survival inequality.

The findings reveal a clear pattern: countries with higher screening coverage achieve earlier diagnosis and substantially better survival outcomes. High-income regions maintain near-perfect survival rates in early stages (~99.5%), while low-income regions start lower (~82%) and experience a steeper decline as the disease progresses.

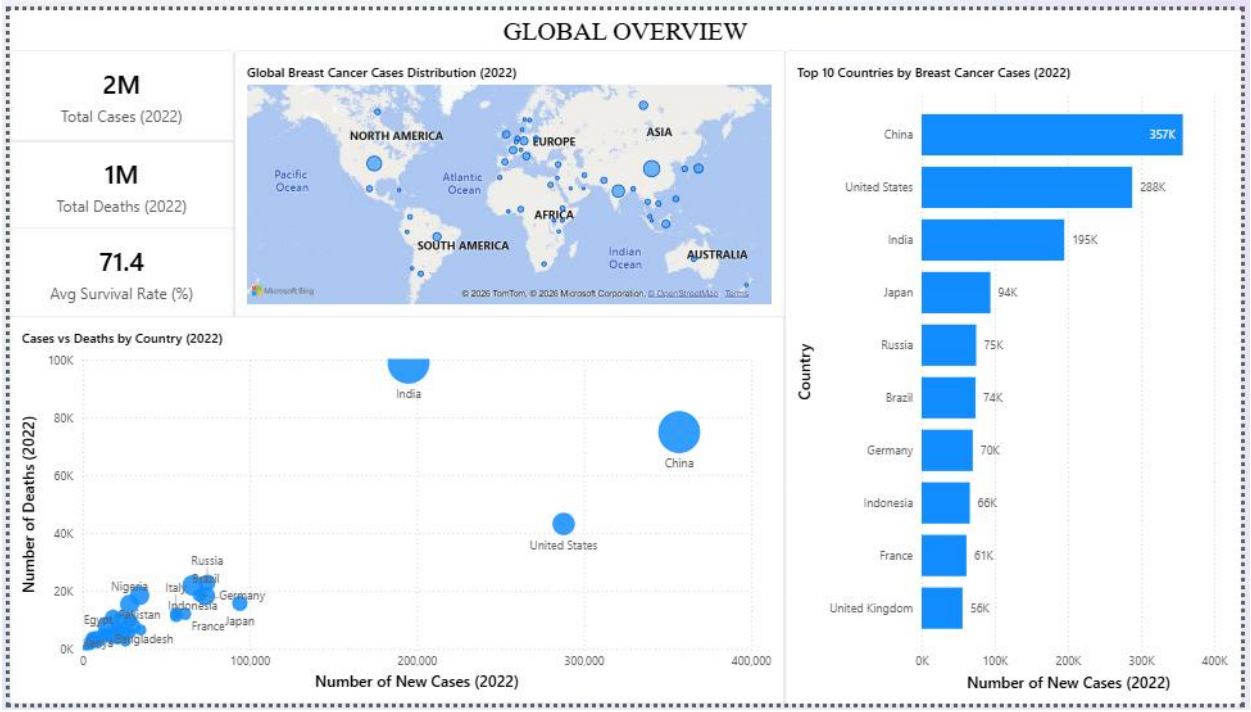
By Stage IV, survival drops to approximately 28% in high-income regions but falls as low as 5% in low-income regions, highlighting a widening gap of over 20 percentage points. This indicates that while early detection is critical, access to advanced treatment becomes increasingly decisive in later stages.

In addition, risk factor analysis shows that a meaningful proportion of breast cancer cases is linked to modifiable lifestyle factors, suggesting that prevention strategies can complement improvements in screening and treatment.

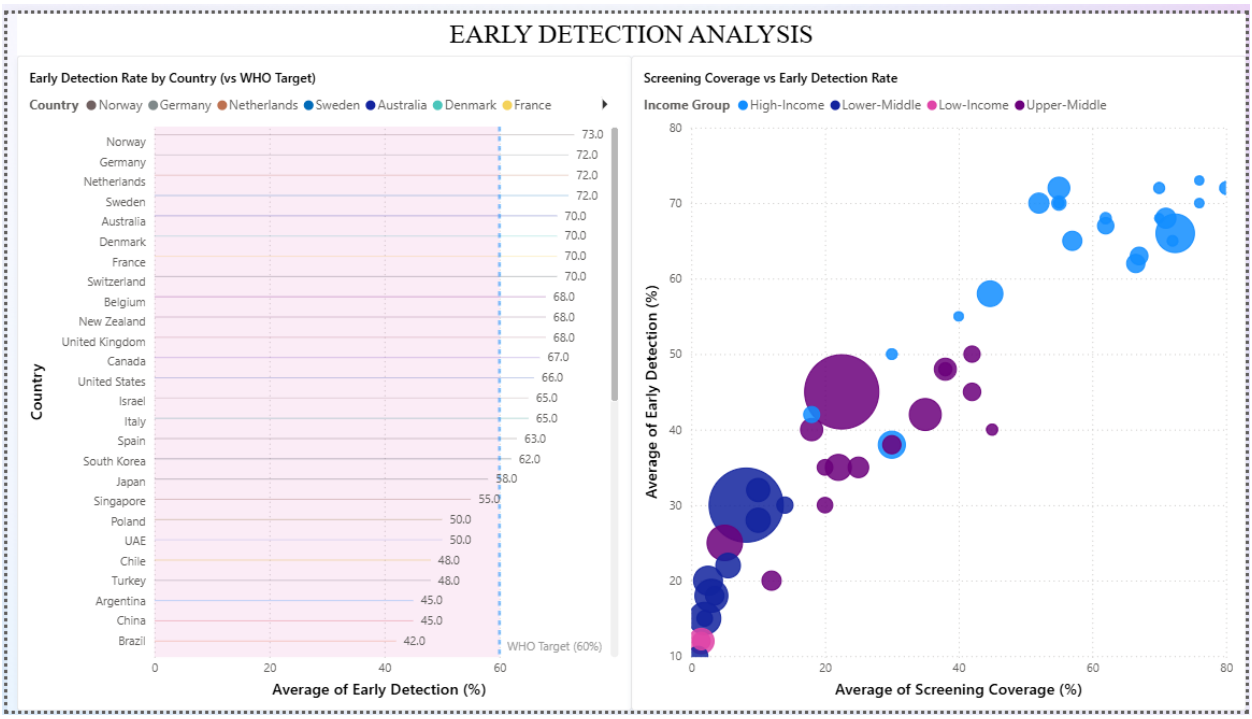
Next Steps:

- Expand screening programs to improve early detection in underserved regions
- Increase access to multi-modal treatment in lower-income countries
- Implement prevention initiatives targeting modifiable risk factors

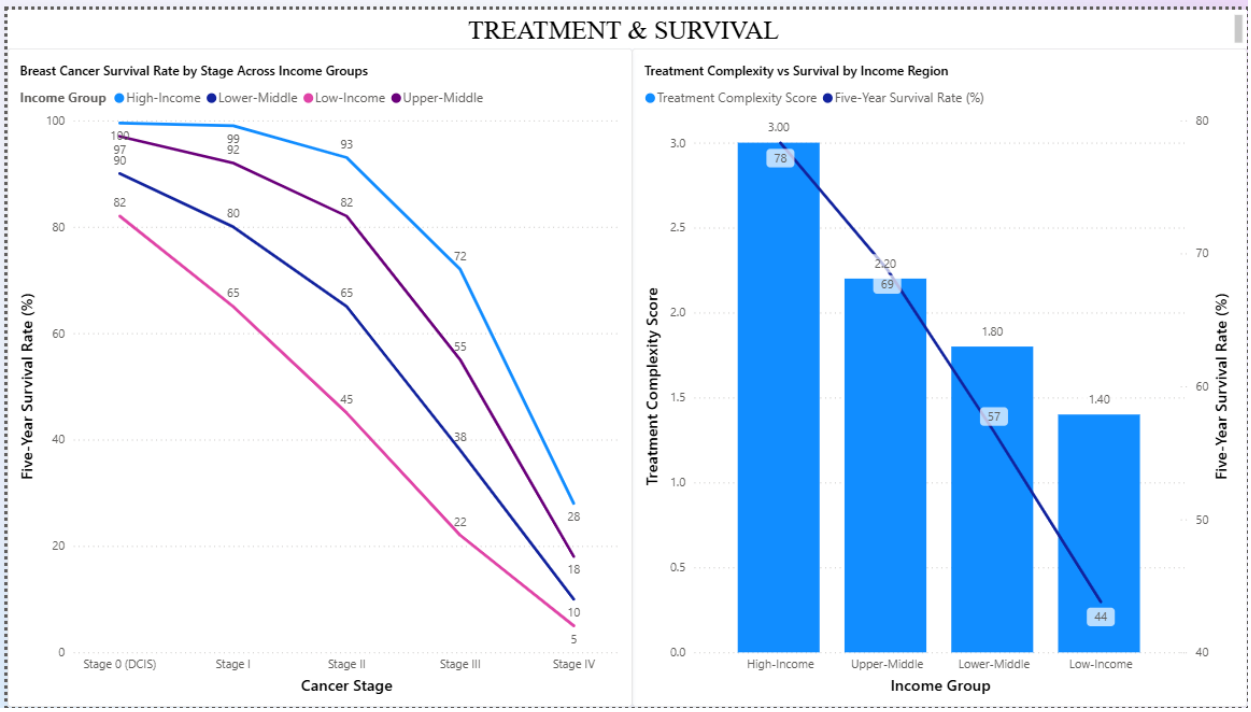
2. Dashboard Preview



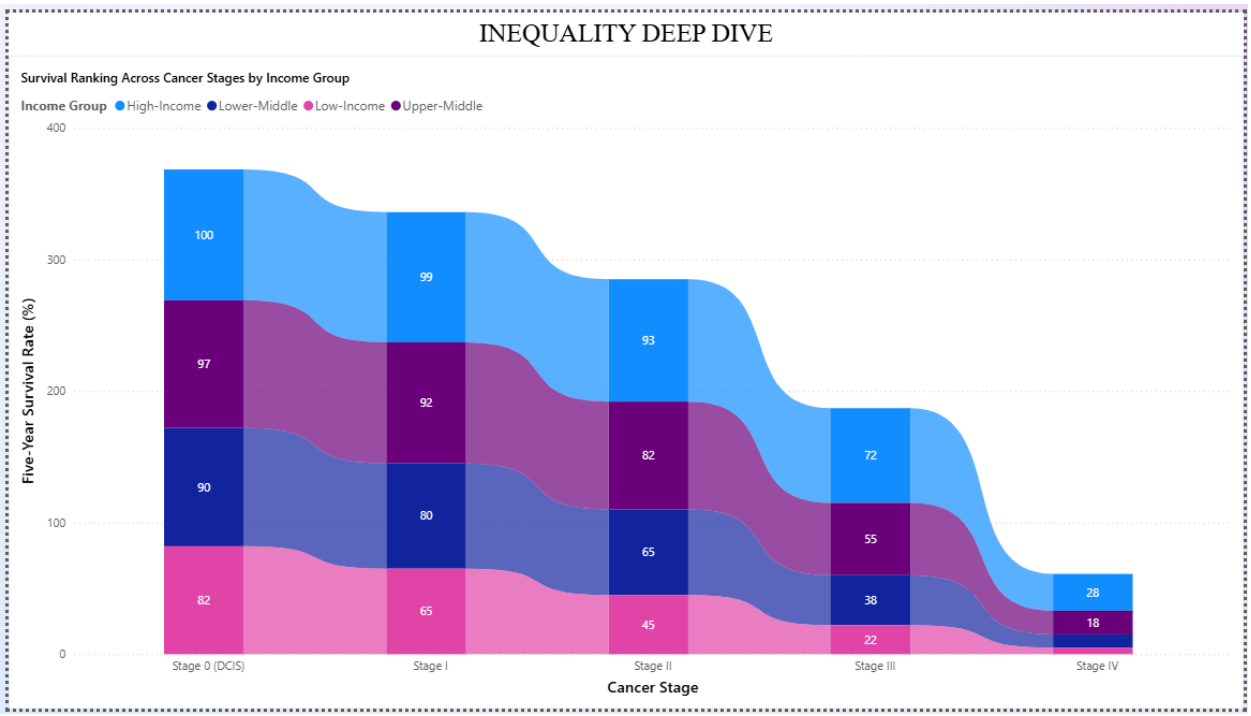
Global distribution of cases and mortality. Highlights top contributing countries.



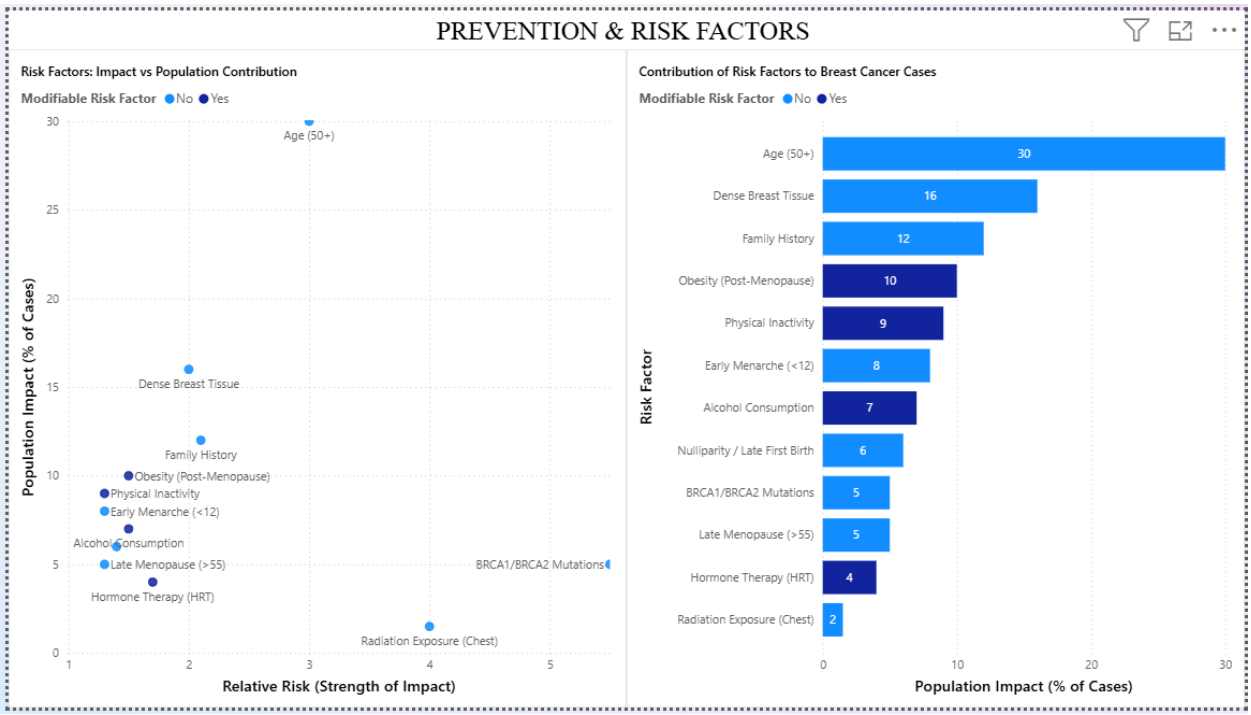
Shows relationship between screening and early detection. Benchmarks countries against WHO targets.



Survival declines across stages. Clear inequality across income groups.



Ranking of survival across stages. Widening gap in later stages.



Identifies high-impact risk factors. Highlights modifiable vs non-modifiable risks.

3. Business Problem

Breast cancer outcomes vary significantly across countries, with survival rates declining sharply in lower-income regions. Despite global improvements in screening and treatment, many countries continue to experience late-stage diagnoses and poor survival outcomes.

Healthcare stakeholders face a critical challenge: understanding how early detection, treatment access, and risk factors interact to influence survival outcomes across different regions.

Without clear insights into these relationships, it becomes difficult to allocate resources effectively, prioritize screening programs, and design interventions that improve survival rates globally.

This project aims to identify the key drivers of survival disparities by analyzing global breast cancer data across detection rates, treatment complexity, and risk factors, with a focus on uncovering actionable opportunities to reduce inequality in outcomes.

4. Methodology

This analysis combines multiple datasets to examine global breast cancer outcomes across detection, treatment, and prevention stages.

- Data Cleaning & Transformation: Used Power Query to clean, standardize, and prepare datasets across countries, stages, and risk factors.
- Data Modeling: Built relationships across tables, including a bridge table for income groups, to enable cross-table analysis.

- Feature Engineering: Created derived variables such as treatment complexity, access level, and detection group classifications.
- Exploratory Data Analysis (EDA): Analyzed global patterns in incidence, mortality, survival rates, and screening coverage.
- Comparative Analysis: Segmented countries by income group to identify disparities in early detection, treatment access, and survival outcomes.
- Stage-Based Analysis: Evaluated survival rates across cancer stages to understand how outcomes evolve as the disease progresses.
- Risk Factor Analysis: Assessed both relative risk and population impact to distinguish between high-impact and high-prevalence risk factors.

5. Skills

Power BI:

- Data modeling (relationships, bridge tables)
- DAX (calculated columns, measures, conditional logic)
- Power Query (data cleaning, transformation, feature engineering)
- Advanced visualizations (map, ribbon chart, scatter plot, combo chart)

Data Analysis:

- Exploratory Data Analysis (EDA)
- Segmentation and cohort analysis (income groups, detection levels)
- Correlation analysis (screening vs detection vs survival)
- KPI development and performance comparison

Analytical Thinking:

- Translating data into business insights
- Identifying inequality and root causes
- Structuring end-to-end analytical narratives

6. Results and Business Recommendations

6.1. Key Insights

Survival outcomes decline significantly as cancer progresses across all income groups. High-income regions maintain near-perfect survival rates in early stages (~99.5%), while low-income regions start lower (~82%) and experience a much steeper decline.

By Stage IV, survival drops to approximately 28% in high-income regions but falls as low as 5% in low-income regions. This represents a widening gap of over 20 percentage points, highlighting increasing inequality in advanced stages.

Early detection varies widely across countries. High-income regions achieve detection rates above 65%, while lower-income regions often remain below 40%, indicating limited access to screening programs.

Treatment access plays a critical role in outcomes. Regions with higher treatment complexity and access consistently achieve better survival rates, particularly in later stages where advanced care is required.

Risk factor analysis shows that while some drivers such as age and genetics are non-modifiable, a meaningful proportion of cases is linked to modifiable lifestyle factors such as obesity, alcohol consumption, and physical inactivity.

6.2. Business Recommendations

- Expand Early Detection Programs
  Invest in screening infrastructure and awareness campaigns in underserved regions to increase early-stage diagnosis rates.
- Improve Treatment Accessibility
  Enhance access to multi-modal treatments (e.g., chemotherapy, radiation, targeted therapy) in lower-income regions to improve survival in advanced stages.
- Prioritize Resource Allocation by Stage Severity
  Focus healthcare investments on later-stage treatment capacity, where survival gaps are most pronounced.
- Strengthen Prevention Strategies
  Implement public health initiatives targeting modifiable risk factors such as obesity, alcohol consumption, and physical inactivity.
- Enable Data-Driven Healthcare Planning
  Leverage data analytics to identify high-risk regions and allocate resources more effectively based on detection and survival gaps.

7. Next Steps

- Incorporate time-series data to analyze trends in detection, treatment, and survival over time.
- Integrate healthcare spending and infrastructure data to assess efficiency and resource allocation.
- Explore patient-level or demographic data for deeper segmentation and predictive modeling.
- Develop forecasting or machine learning models to predict survival outcomes based on detection and treatment variables.
- Enhance geographic granularity (e.g., regional or city-level analysis) for more targeted policy recommendations.